

From Bins to Bell Curve

How a handful of discrete Mendelian loci add up to a continuous trait — and what heritability does and does not tell you once environment is in the picture.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/bins-to-bell-curve/

1 Before you open anything

Answer from what you already know. Being right is not the point — having something on paper to argue with later is.

PREDICTION

In this model, one locus ($Aa \times Aa$) produces three possible offspring heights spread across 150–190 cm. Now suppose **eight** unlinked loci share the job, each capital allele adding the same amount, and the trait still spans 150–190 cm. With no environmental effect at all, how many *distinct* heights can the offspring have?

Write one sentence saying how you got that number.

2 Reading the cross

Open the demonstration. The **The cross** panel sets up the parents; the **Genetics alone** panel reports what that cross produces before environment is involved.

1. Press **Reset**. The loci slider should read **1** and the environmental slider **0.00 cm**.
2. From the **Genetics alone** panel, copy **Phenotype classes ($2n+1$)**, **Gap between classes**, **Offspring in one extreme class** and **Class ratio** into the first row below.
3. Drag **Contributing loci**, n to **2**, then to **3**, filling in a row each time.

n	Phenotype classes	Gap between classes (cm)	Offspring in one extreme class	Class ratio
1				
2				
3				

a. The class ratio at $n = 1$ is one you already know from a monohybrid cross. Write the genotypes it corresponds to, in order from 150 cm to 190 cm.

b. At $n = 2$ the ratio has five numbers instead of three. Where does each number come from? (You are counting allele combinations across four alleles, not across two.)

c. Look at the plot at $n = 3$. The caption calls the bars *point masses*. Using your gap value, state one height between 150 and 190 cm that **no** offspring in this model can have, and say why.

The sliders move in fixed steps — loci in steps of 1, environmental SD in steps of 0.25 cm. If dragging overshoots, click the slider once and use the arrow keys to land on an exact value. The number beside each label always shows the setting you are actually on.

3 What actually turns bars into a bell

Keep **Environmental SD, σ_E** at **0.00 cm** for this whole part. Nothing here is about environment yet.

1. Use the **$n = 1$** button, then the loci slider for 2 and 4, then the **$n = 8$** button.
2. At each setting record the three values from **Genetics alone** plus **Genetic SD** from the **Variance in this population** panel.
3. Watch the plot, not just the numbers, as you go from 1 to 8.

n	Phenotype classes	Gap between classes (cm)	Offspring in one extreme class	Genetic SD (cm)
1				
2				
4				
8				

a. Write the number of classes and the gap between classes as formulas in n .

classes = _____ gap = _____ cm

b. Divide the $n = 2$ extreme-class value by the $n = 1$ value. What factor does one additional locus multiply that frequency by, and why that factor and not some other one? Then read the line under the $n = 8$ value: one offspring in how many?

c. Three things change together as n rises, and a bell needs all three. Name them, in two sentences, using your table.

d. Go back to your prediction in part 1. Compare it with the $n = 8$ row. If the numbers differ, say specifically what you assumed that the model does not do.

e. Genetic SD *falls* from 14.14 cm to 5.00 cm as loci are added, even though every offspring is still genetically determined. Explain this using the fact that the 150–190 cm range is held fixed no matter how many loci share it.

4 Reading it backwards, and one shortcut that fails

Exam questions rarely hand you n . They hand you something the distribution does, and expect n back.

1. A cross of this kind produces 11 evenly spaced phenotype classes, 4.00 cm apart, across 150–190 cm. Work out n on paper first, then set the slider and check.

a. $n =$ _____ Which of the two clues did you use, and would the other one alone have been enough?

Now a claim worth testing: *more loci contributing to a trait means more of its variation is genetic, so heritability goes up*. It sounds reasonable. Test it.

1. Set σ_E to **5.00 cm** and leave it there for both rows.
2. Record the four values at **$n = 1$** , then at **$n = 8$** . V_G and V_E are printed under the split bar at the bottom of the **Variance in this population** panel.

Setting	V_G (cm ²)	V_E (cm ²)	Total SD (cm)	Heritability h^2
$n = 1, \sigma_E = 5.00$				
$n = 8, \sigma_E = 5.00$				

b. Which way did h^2 move? Since σ_E never changed, the only term that moved was V_G . Say what happened to it and why, and then state what the claim above got wrong.

c. That result depends on a modelling choice, not on biology: adding loci here divides a fixed 150–190 cm range into finer steps rather than widening the range. Read the **What the model assumes** panel and name one other simplification that this demonstration makes.

d. With $n = 2$, find the setting of σ_E that makes h^2 read exactly 0.50. Then, **before** touching the slider, predict the value that would do the same at $n = 8$, and check it.

σ_E at $n = 2$: _____ cm predicted at $n = 8$: _____ cm actual:
_____ cm

Watch the wording. Every h^2 on this page describes the variation among *this set of offspring*. It never says that some fraction of one individual's height came from their genes. That second question has no answer, because no part of anyone's height was built without both.

5 Solve these without the demonstration

Close the page or look away. Show your working. Then reopen it, check yourself, and if a number is off, write down which step produced it.

1. Three loci contribute additively to a trait spanning 150–190 cm, all parents fully heterozygous, no environmental effect. Give the number of phenotype classes, the gap between neighbouring classes, and the fraction of offspring with the all-lowercase genotype.

2. Five loci, same trait, with an environmental SD of 6.00 cm. Find V_G , V_E , the total SD, and h^2 to two decimal places. (Genetic variance in this model is $200/n \text{ cm}^2$.)

3. Population A: one locus, no environmental variance. Population B: eight loci, $\sigma_E = 12.00$ cm. Which population's offspring are more spread out in height? Compute both total SDs before answering, then say why comparing total SD across two populations tells you nothing on its own about which one experiences more environmental variance.

4. No calculation for this one. A study of a plant population reports $h^2 = 0.53$ for stem height. Explain what that number does say, and state one conclusion people commonly draw from a heritability estimate that it does not support. Then name one condition that has to hold for the estimate to describe any population other than the one measured.

6 On your own

Nothing here is graded, and the worksheet has deliberately left these alone.

- The **Overlay normal approximation** checkbox stays locked below $n = 4$. Turn it on at $n = 4$ and again at $n = 8$ and find where the dashed curve sits above or below the bars. The bars are the exact binomial; the curve is only an approximation to it.
 - The dots along the bottom are 260 sampled offspring, and **New sample** draws a fresh set. At $n = 8$ with $\sigma_E = 0$, how many samples do you get through before one offspring lands in an extreme class? Compare that with the frequency the panel reports.
 - Set $n = 2$ and raise σ_E one step at a time from 0.00. Find the setting at which the caption stops describing the classes as separate humps, and check it against half the gap between classes. Then explain why *half* the gap is the sensible place for that boundary.
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